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SASexp3a.m | SASexp4a.m

```
1   % Experiment 11: Linearity and Time Invariance
2 - clc; clear; close all;
3 - t = -1:0.001:1;
4 - x1 = sin(2*pi*5*t); x2 = cos(2*pi*5*t);
5 - a = 2; b = 3; t0 = 0.2;
6 - T = @(x) 2*x;
7 - lhs = T(a*x1+b*x2);
8 - rhs = a*T(x1)+b*T(x2);
9 - linError = max(abs(lhs-rhs));
10 - xDelay = sin(2*pi*5*(t-t0));
11 - lhsTI = T(xDelay);
12 - rhsTI = T(x1);
13 - rhsTIshift = interp1(t,rhsTI,t-t0,'linear',0);
14 - tiError = max(abs(lhsTI-rhsTIshift));
15 - figure;
16 - subplot(2,1,1); plot(t,lhs-rhs); grid on; title('Linearity Error');
17 - subplot(2,1,2); plot(t,lhsTI-rhsTIshift); grid on; title('Time-Invariance Error');
18 - fprintf('Linearity error = %g\n',linError);
19 - fprintf('Time-invariance error = %g\n',tiError);
```

Workspace

Name	Value
a	2
b	3
lhs	*1x2001 double*

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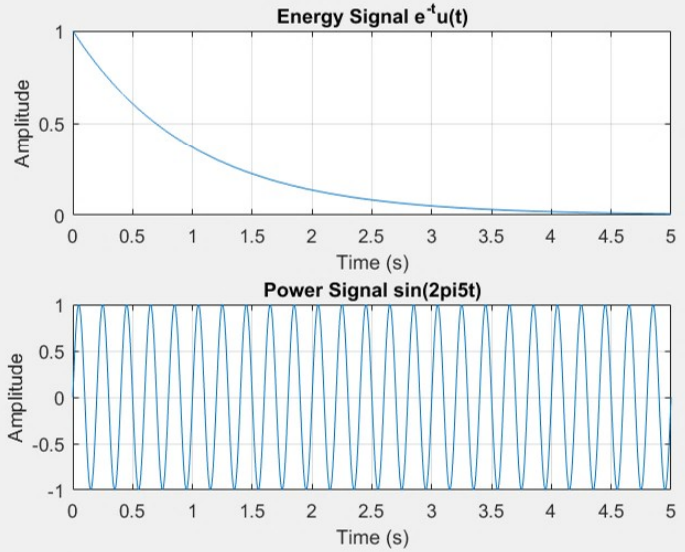
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SASexp3a.m

```
1   % Experiment 3: Energy and Power Analysis
2   clc; clear; close all;
3
4   dt = 0.001;
5   t = 0:dt:5;
6
7   xE = exp(-t);
8   xP = sin(2*pi*5*t);
9
10  E_num = trapz(t,abs(xE).^2);
11
12  T0 = 1/5;
13  idx = t <= 2;
14  P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
15      trapz(t(idx),abs(xP(idx)).^2);
16
17  figure;
18  subplot(2,1,1);
19  plot(t,xE); grid on;
20  title('Energy Signal e^{-t}u(t)');
21  xlabel('Time (s)'); ylabel('Amplitude');
22
23  subplot(2,1,2);
24  plot(t,xP); grid on;
25  title('Power Signal sin(2pi5t)');
```

Figure 1

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Workspace

Name	Value
dt	1.0000e-03
E_num	0.5000
idx	1x5001 logical

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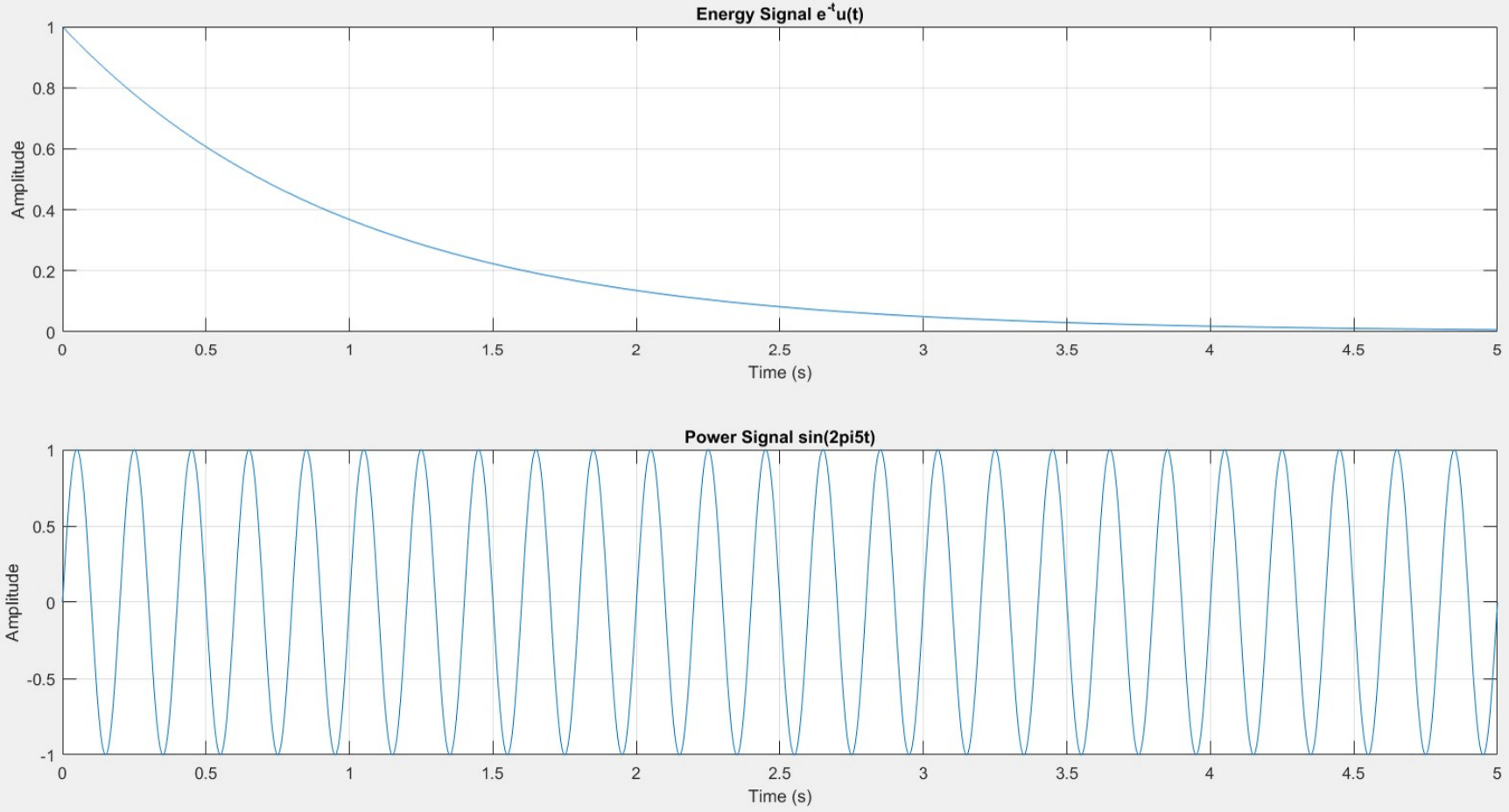
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Figure 1

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Command Window

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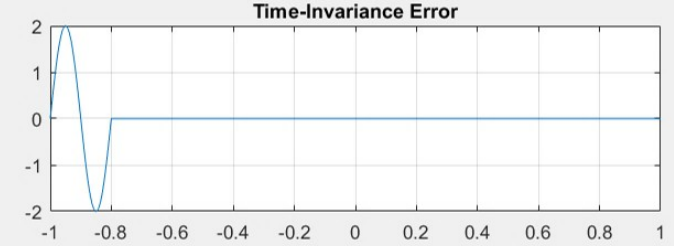
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Figure 1

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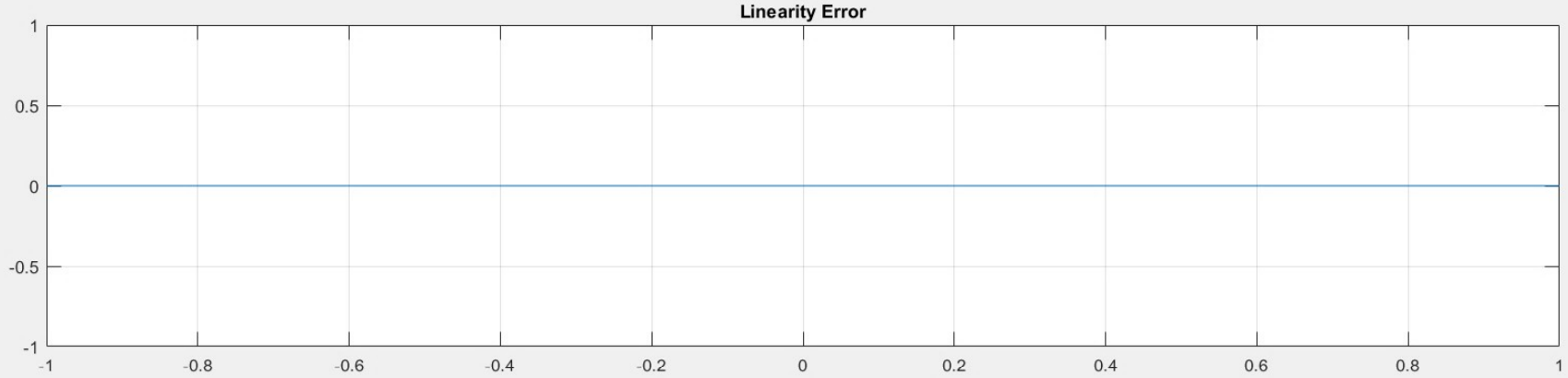
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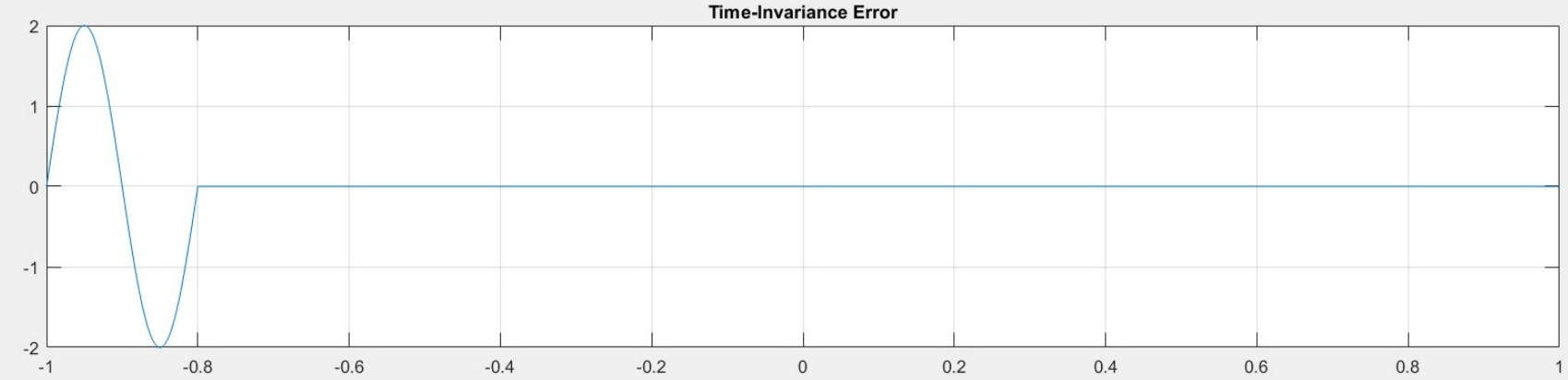
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Figure 1

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